

Complex Number 15

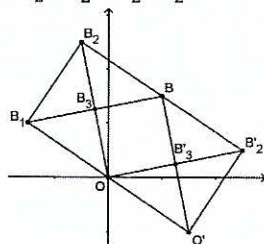
- z has modulus r and argument θ . Find the modulus and argument in terms of r and θ
 - z^2
 - $\frac{1}{z}$
 - iz
- $(1 + \sqrt{3}i)^{-1}$ in modulus/argument form.
 - State the modulus and argument of $-1 + i$. Hence write $(-1 + i)^{18}$ in the form $a + ib$.
- Write $\sqrt{3} + i$ and $\sqrt{3} - i$ in modulus/argument form. Hence write $(\sqrt{3} + i)^{10} + (\sqrt{3} - i)^{10}$ in the form $a + ib$.
- Find the modulus of $\frac{7-i}{3-4i}$. Evaluate $\tan \left\{ \tan^{-1} \left(\frac{4}{3} \right) - \tan^{-1} \left(\frac{1}{7} \right) \right\}$. Hence find the principal argument of $\frac{7-i}{3-4i}$ in terms of π .
- Find a and b such that $(1 + i\sqrt{3})^8 = a + ib$.
- Find $\sqrt{6i - 8}$ and hence solve the quadratic equation: $2z^2 - (3 + i)z + 2 = 0$ expressing your answer in the form $x + iy$.
- If ω is a complex root of the equation: $z^5 - 1 = 0$.
 - Show that $\omega^2, \omega^3, \omega^4$ are the other complex roots.
 - Prove that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$
 - Form the quadratic equation with roots $(\omega^2 + \omega^3)$ and $(\omega + \omega^4)$.
- If $z = \cos \theta + i \sin \theta$ show that: $z + z^{-1} = 2 \cos \theta$ and $z^n + z^{-n} = 2 \cos n\theta$ and find corresponding expressions for $z - z^{-1}$ and $z^n - z^{-n}$.
 - By expanding $\left(z + \frac{1}{z}\right)^4$ find an expression for $\cos^4 \theta$ in the form $A \cos 4\theta + B \cos 2\theta + C$.
- By considering the triangle inequality $|z_1 - z_2| \geq |z_1| - |z_2|$ find the greatest value of $|z|$ which is satisfied by the equation $\left|z - \frac{7}{z}\right| = 4$.
- If z is a complex number such that $z = k(\cos \theta + i \sin \theta)$, where k is real, show that $\arg(z + k) = \frac{\theta}{2}$.
- By solving the equation $z^3 = 1$ find the three cube roots of one.
 - Let ω be a cube root of one where ω is not real. Show that $1 + \omega + \omega^2 = 0$.
 - Given $x = a + b, y = a\omega + b\omega^2$ and $z = a\omega^2 + b\omega$, prove that $x^2 + y^2 + z^2 = 6ab$
- Show that $\frac{(1+i\sqrt{3})^6}{(\sqrt{3}-i)^k} = 2^{6-k} \left(\cos \frac{k\pi}{6} + i \sin \frac{k\pi}{6} \right)$
 - For what value of k is $\frac{(1+i\sqrt{3})^6}{(\sqrt{3}-i)^k}$ purely imaginary.
- Let $OABC$ be a square in the Argand diagram where O is the origin. The points A and C represent the complex numbers z and iz respectively.
 - Find the complex number represented by B .
 - The square $OABC$ is now rotated through 45° in an anticlockwise direction about O to $OA'B'C'$. Find the complex numbers A', B' and C' .
- P, Q and R represent the complex numbers z_1, z_2 and z_3 respectively on the Argand diagram. Given that $z_1 - z_2 = i(z_2 - z_3)$ show this information on a diagram and hence fully describe ΔPQR .
- The point P on the Argand diagram represents the complex number z where z satisfies the equation: $|z - 2i| = \operatorname{Im}(z + 2i)$. Find the Cartesian equation of the locus of P and give a geometric description of the locus of P .

16. By applying the De Moivre's Theorem, and by also expanding $(\cos \theta + i \sin \theta)^4$ express $\tan 4\theta$ as a polynomial in terms of $\tan \theta$.
17. A represents the complex number $2 + 3i$ in the complex plane. Find a possible complex number represented by B such that triangle OAB is a right angled isosceles triangle with right angle at
- O
 - A
 - B
18. Simplify each expression, leaving your answer in mod-arg form:
- $5\text{cis}\frac{\pi}{12} \times 2\text{cis}\frac{\pi}{4}$
 - $3\text{cis}\theta \times 3\text{cis}2\theta$
 - $6\text{cis}\frac{\pi}{2} \div 3\text{cis}\frac{\pi}{6}$
 - $\frac{3\text{cis}5\alpha}{2\text{cis}4\alpha}$
 - $\left(4\text{cis}\frac{\pi}{5}\right)^2$
 - $\left(2\text{cis}\frac{2\pi}{7}\right)^3$
19. In the complex plane, mark a point K to represent a complex number $z = r \text{cis}\theta$ where $1 < r < 2$ and $\frac{\pi}{4} < \theta < \frac{\pi}{2}$. Hence indicate clearly the points M, N, P, Q and R representing $\bar{z}, -z, 2z, iz$ and $\frac{1}{z}$ respectively.
20. Let z be a non-zero complex number such that $0 < \arg z < \frac{\pi}{2}$. Indicate points A, B, C and D in the complex plane representing the complex number $z, -iz, \left(2 \text{cis}\frac{\pi}{3}\right)z$ and $\left(\frac{1}{2} \text{cis}\left(-\frac{\pi}{4}\right)\right)z$
21. Given that $z_1 = \sqrt{3} + i$ and $z_2 = 2\sqrt{2} + 2\sqrt{2}i$
- Write z_1 and z_2 in mod-arg form;
 - Hence write $z_1 z_2$ and $\frac{z_2}{z_1}$ in mod-arg form.
22. Repeat the previous question for $z_1 = -\sqrt{3} + i$ and $z_2 = -1 - i$

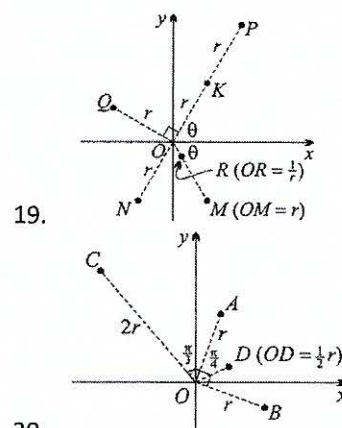
Answers

- $|z^2| = r^2, \arg(z^2) = 2\theta$
 - $\left|\frac{1}{z}\right| = \frac{1}{r}, \arg\left(\frac{1}{z}\right) = -\theta$
 - $|iz| = r, \arg(iz) = \frac{\pi}{2} + \theta$
- $\frac{1}{2} \text{cis}\left(-\frac{\pi}{3}\right), \sqrt{2} \text{cis}\frac{3\pi}{4}, -512i$
- $2 \text{cis}\frac{\pi}{6}, 2 \text{cis}\left(-\frac{\pi}{6}\right), 1024$
- $\sqrt{2}; 1, \frac{\pi}{4}$
- $a = -128, b = 128\sqrt{3}$
- $z = (1 + i) \text{ or } \frac{1}{2}(1 - i)$
- $c. z^2 + z - 1 = 0$
- $z - z^{-1} = 2i \sin \theta, z^n - z^{-n} = 2i \sin n\theta$
 - $A = \frac{1}{8}, B = \frac{1}{2}, C = \frac{3}{8}$
- $\max|z| = 2 + \sqrt{11}$
- $a. z = 1, \frac{-1+\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}$
- $k = 3 + 6n\pi, n = 0, \pm 1, \pm 2, \dots$
- $a. z_2 = (1 + i)z, A' \rightarrow z'_1 = \frac{z}{\sqrt{2}}(1 + i), B' \rightarrow z'_2 = iz\sqrt{2}, C' \rightarrow z'_3 = \frac{z}{\sqrt{2}}(-1 + i)$

- ΔPQR is an isosceles right angled triangle with right angle at $Q (z_2)$
- $x^2 = 8y$. A parabola: Vertex O and focus $(0, 2)$
- $3 - 2i$
 - $-1 + 5i$ or $5 + i$
 - $-\frac{1}{2} + \frac{5i}{2}$ or $\frac{5}{2} + \frac{i}{2}$



- $10\text{cis}\left(\frac{\pi}{3}\right)$
 - $9\text{cis}3\theta$
 - $2\text{cis}\frac{\pi}{3}$
 - $\frac{3}{2} \text{cis} \alpha$
 - $16\text{cis}\frac{2\pi}{5}$
 - $8\text{cis}\frac{6\pi}{7}$



- $z_1 = 2\text{cis}\frac{\pi}{6}$ and $z_2 = 4\text{cis}\frac{\pi}{4}$
 - $z_1 z_2 = 8\text{cis}\frac{5\pi}{12}$ and $\frac{z_2}{z_1} = 2\text{cis}\frac{\pi}{12}$
- $z_1 = 2\text{cis}\frac{5\pi}{6}, z_2 = \sqrt{2}\text{cis}\left(-\frac{3\pi}{4}\right), z_1 z_2 = 2\sqrt{2}\text{cis}\frac{\pi}{12}$ and $\frac{z_2}{z_1} = \frac{\sqrt{2}}{2} \text{cis}\frac{5\pi}{12}$